A Better Day at the Park: Attraction Selection under Changing Queues and Road Conditions

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Abstract

A visitor with limited time must balance attraction value against walking, queueing and the need to reach an exit. I compare distance-based routes, an adaptive heuristic, Q-learning and a deep Q-network (DQN) on a ten-attraction park map. Four scenarios vary crowd levels and disruption timing. The DQN achieves the highest mean reward in each scenario, although its gains over adaptive greedy selection are modest. A shorter-budget experiment reveals weaker deadline performance. These results connect route quality to current conditions, visitor preferences and the reliability of time estimates.

1 Introduction

A map makes the shortest route easy to imagine, but a busy attraction can consume much of a day in a queue. A visitor may also prefer a longer walk to a more interesting attraction, or change plans when a road closes. These choices raise a practical question: how should the next destination depend on both the remaining time and the changing environment?

I study this question through a visitor who chooses attractions one at a time. The analysis asks whether learned decisions collect more attraction value than distance-based routes, how those gains compare with a simple adaptive heuristic, and whether the policies remain effective when a disruption changes timing or the time budget becomes shorter. The comparison includes both trip quality and successful exit, because a rewarding sequence of visits can still exceed the available time.

Earlier itinerary research has incorporated queueing time and personal interests [2], while crowd-aware recommendation has considered interactions between route choices and aggregate welfare [3]. The present analysis focuses on one visitor facing externally changing conditions. Q-learning [4] and DQN [5] provide two ways to learn the consequences of successive attraction choices.

2 Public Attraction Data

The input is the Magic Kingdom attraction table published in the SCAIR repository [1]. The downloaded CSV contains 27 records and eight fields: identifier, name, latitude, longitude, duration, capacity, rating and review count. The file contains no missing values or duplicate identifiers. Its ratings range from 3.2 to 4.7, and its duration field ranges from 1.5 to 30. The analysis uses ten specified attractions, listed in Table 1. The source is a historical snapshot; its entries do not establish current operating conditions.

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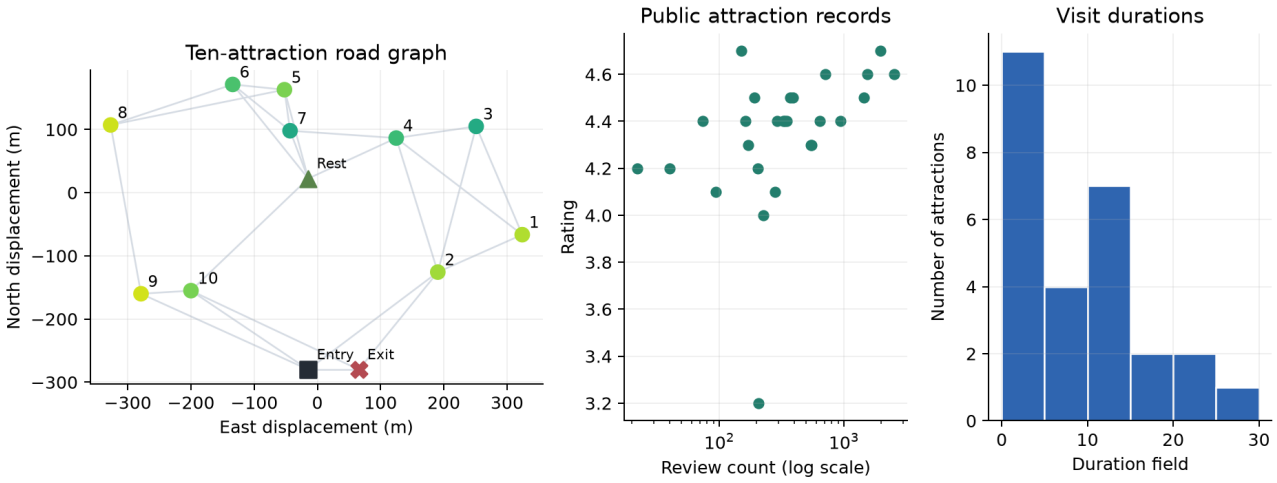


Figure 1: The selected attraction map and descriptive distributions from all 27 public records. Node numbers match Table 1; node colours indicate relative base value. The duration field is treated as minutes in the simulation.

3 Simulation Analysis

3.1 Map, values and available actions

The map contains ten attractions, an entry, an exit and a rest node. A local projection converts coordinates to metres. Three-nearest-neighbour connections and a minimum spanning tree give 25 undirected roads. The entry and exit lie south of the attractions; the rest node lies at their centroid.

For attraction j, the base value is

$$b_j = \tfrac{1}{2}\frac{r_j}{5} + \tfrac{1}{2}\frac{\log(1+n_j)}{\log(1+n_{\max})}, \qquad u_{pj} = b_j \eta_{pj}, \quad \eta_{pj} \sim U(0.7, 1.3), \tag{1}$$

where r_j is the rating, n_j is the review count and $n_{\max}$ is the full source table's largest count. Each case selects one of three fixed preference profiles. The environment treats durations as minutes, although the CSV does not specify their units. Capacity does not enter the queue model.

Table 1: Selected attractions and their public source attributes.

Map label	Attraction	Duration	Rating	Reviews
1	Space Mountain	10	4.6	1548
2	Buzz Lightyear's Space Ranger Spin	4.5	4.5	1449
3	Dumbo The Flying Elephant	1.5	4.4	163
4	The Many Adventures of Winnie the Pooh	4	4.4	326
5	It's A Small World	14	4.6	706
6	Haunted Mansion	10	4.5	372
7	Peter Pan's Flight	3	4.2	203
8	Big Thunder Mountain Railroad	7	4.7	1956
9	Pirates of the Caribbean	7.5	4.6	2547
10	Jungle Cruise	8	4.4	934

An action selects an attraction, exits, or takes an eight-minute break at the rest node. A transfer follows a shortest path at departure-time road costs, followed by the arrival-time queue and visit duration. Completion after closing yields no attraction value. Reaching the exit or exceeding the budget ends the episode.

3.2 Queues, roads and disruptions

The simulation defines queues, roads, opening windows and preferences. The base queue is $4 + 13p_j$ minutes, where p_j is normalized log review count. A sinusoidal term and Gaussian noise with standard deviation 1.8 minutes vary queues in ten-minute noise intervals. Road costs receive independent multipliers in $[0.95, 1.70]$ every fifteen minutes relative to a walking speed of 70 metres per minute.

The calm scenario uses background changes. Peak conditions multiply queues by 1.65 and road costs by 1.30. The disruption adds 22 queue minutes at the three highest-value attractions, multiplies road costs by 1.35 and closes a frequently used interior road from minute 60 to 115. The graph stays connected. The shifted event starts at minute 40, lasts 55 minutes and adds 32 queue minutes. Two attractions open at minutes 20 and 35; two close at minutes 115 and 125. Others close at minute 220.

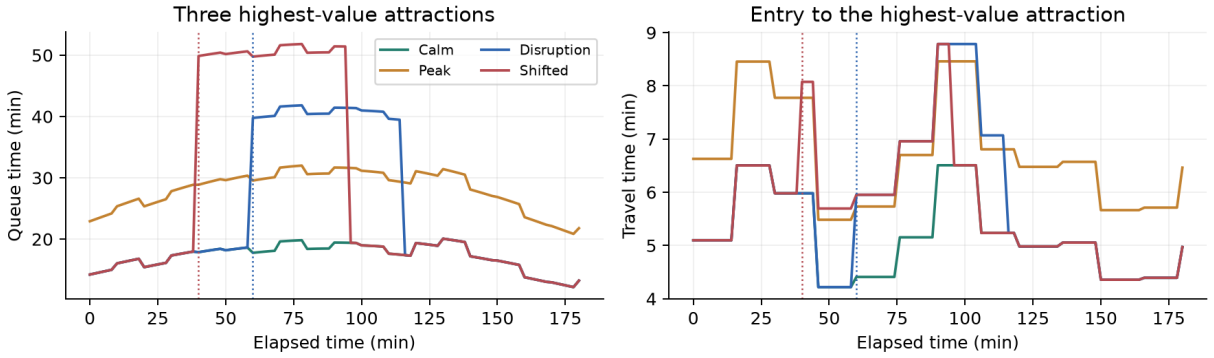


Figure 2: Queue and travel changes for a fixed case. The dotted lines mark the disruption onsets at minutes 40 and 60. The road-cost snapshot remains fixed during each transfer; queues update on arrival.

4 Decision Methods

4.1 Observations, feasibility and reward

Each decision observes the current location, visited attractions, elapsed time, remaining-time fraction, preference profile and attraction values. The DQN also receives current queues, travel times to attractions, estimated attraction-to-exit times, opening status and whether the rest action has been used. These components form a 79-dimensional vector. The visitor observes current conditions without receiving the future queue or road trace.

All policies use the same action mask. An attraction must be unvisited and currently open. Its estimated travel, queue and visit time must permit completion before closing and leave enough time for an estimated transfer to the exit:

$$t + \widehat{w}_{ij}(t) + q_j(t) + d_j + \widehat{w}_{jE}(t) \le B. \tag{2}$$

Here B is the budget, d_j is the attraction duration, and $\widehat{w}$ is a shortest-path travel estimate under current road conditions. The mask can become optimistic when conditions change before arrival. The exit action remains available, and the rest action is allowed at most once when its estimated cost leaves time to exit.

The immediate reward is

$$R = 10u_{pj}I_j - 0.08w - 0.12q - 0.025(d+h) + z, \tag{3}$$

where I_j indicates a successfully completed visit, w is walking time, q is queue time, d is visit time and h is rest time. A timely exit gives $z = 5$; a late exit or budget overrun gives $z = -25$. Other actions have $z = 0$. Reported reward is the undiscounted episode sum, while training uses discount factor $\gamma = 0.98$.

4.2 Distance-based and adaptive baselines

The *frozen route* fixes a nearest-neighbour attraction order using the initial road network and selects the first currently feasible attraction in that order. The *nearest* policy recomputes shortest-path distances at each decision and chooses the nearest feasible attraction. Both policies leave when no attraction remains feasible. The *adaptive greedy* policy instead maximizes

$$\frac{u_{pj}}{\widehat{w}_{ij}(t) + q_j(t) + d_j} \tag{4}$$

among feasible attractions. This heuristic directly accounts for current value and time cost, so it provides a stronger comparison than distance alone.

4.3 Q-learning and DQN

Q-learning updates a table through the temporal-difference target $R + \gamma \max_{a'} Q(s', a')$. Its aggregated key contains location, a visited-attraction bit mask, twenty-minute elapsed-time bins, preference profile, mean-queue bins, rest status, current disruption status and thirty-minute budget bins. Unseen keys start with zero values. This aggregation omits the full queue and road vectors, so the comparison involves both learning and state representation.

The DQN approximates action values with two 48-unit ReLU hidden layers and twelve outputs. It uses a 12,000-transition replay buffer, batches of 64, Huber loss and Adam with learning rate 10^{-3}. A separate target network updates every 200 gradient steps. Training takes one gradient step every two transitions after a 256-transition warm-up. Both learners use an ϵ-greedy schedule that decreases from 1 to 0.06. Each algorithm trains for 2,500 episodes per seed under a mixture of calm, peak and disruption conditions.

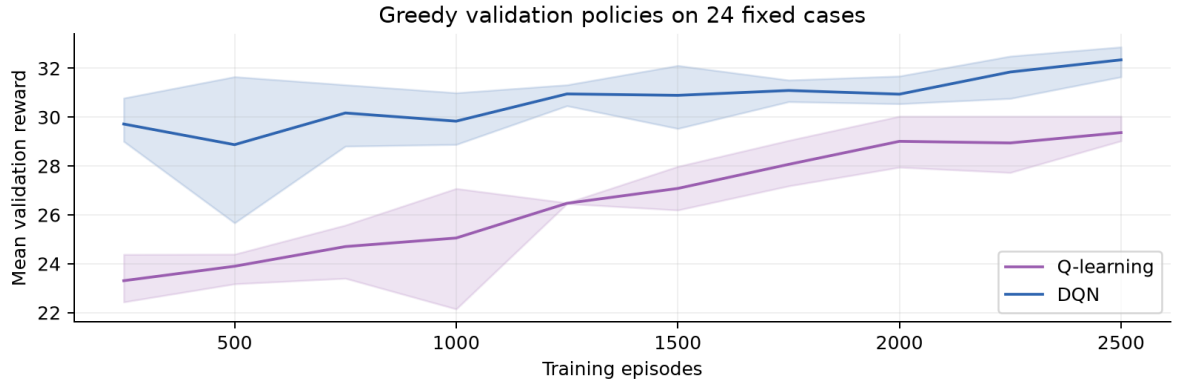


Figure 3: Mean validation reward across three training seeds. Shading spans the smallest and largest seed means. The 24 fixed validation cases are separate from training and testing.

5 Evaluation and Main Results

The main experiment uses an initial budget of 150 minutes and 80 cases per scenario. Every policy receives the same preference profile, queue noise and road-cost trace for a given case. The learners use three independent training seeds, and the final model from each seed enters evaluation without further learning. The shifted disruption appears only in testing. Validation monitors progress without selecting a checkpoint.

Attraction coverage is $100|V|/10$, where V is the set of successfully visited attractions. Preference satisfaction is $100 \sum_{j \in V} u_{pj} / \sum_j u_{pj}$; this score describes collected model value rather than a measured survey response. On-time completion requires reaching the exit within budget. Walking and queue time measure separate components of the trip, and reward combines value, time costs and completion.

Table 2 reports case means, averaged across the three trained models for learned policies. Deterministic baseline results repeat across the training-seed slots solely to align comparisons; these copies do not increase the number of independent cases. The stale-input DQN freezes its queue and travel features at their initial values but retains current opening information and the shared feasibility mask. This inference-time ablation tests the contribution of live ranking features; the ablation also introduces an input shift relative to training.

Table 2: Mean outcomes with a 150-minute budget. Coverage, preference satisfaction and completion are percentages; queueing and walking are minutes.

Scenario	Policy	Reward	Coverage	Preference	Queue	Walk	Complete
Calm	Frozen route	35.5	51.5	50.8	78.4	28.2	97.5
	Nearest	35.6	52.1	51.4	78.9	27.8	96.2
	Adaptive greedy	38.1	51.6	54.4	75.2	33.9	95.0
	Q-learning	35.2	46.5	49.0	71.1	35.6	100.0
	DQN	39.7	51.0	55.0	77.8	30.2	98.8
	DQN stale input	39.4	50.6	54.4	77.1	31.6	99.6
Peak	Frozen route	19.3	32.2	32.5	84.1	29.3	97.5
	Nearest	19.2	32.4	32.7	83.6	28.9	96.2
	Adaptive greedy	22.9	32.6	35.8	78.7	31.4	97.5
	Q-learning	21.3	30.0	33.4	78.0	32.7	100.0
	DQN	24.4	36.4	38.8	89.2	30.2	97.9
	DQN stale input	23.9	35.3	38.2	86.9	32.2	97.5
Disruption	Frozen route	33.5	48.9	48.2	75.5	30.9	97.5
	Nearest	34.0	49.5	48.8	75.1	30.9	97.5
	Adaptive greedy	37.1	49.8	51.2	72.4	35.5	100.0
	Q-learning	33.0	43.2	46.1	66.9	37.9	99.6
	DQN	38.1	49.8	53.4	76.2	33.9	98.3
	DQN stale input	37.6	49.1	52.9	75.0	36.0	98.3
Shifted	Frozen route	24.2	42.5	42.0	85.1	28.9	87.5
	Nearest	24.3	43.0	42.4	84.1	29.4	86.2
	Adaptive greedy	36.8	50.9	51.6	73.7	36.5	98.8
	Q-learning	26.5	36.1	38.0	61.1	35.1	97.9
	DQN	37.9	49.9	53.1	75.3	33.7	98.3
	DQN stale input	36.4	48.0	51.8	76.0	35.5	97.5

The DQN has the highest mean reward in every scenario. Its reward exceeds the live nearest policy by 4.02, 5.16, 4.07, 13.56 points in calm, peak, disruption and shifted conditions, respectively. The largest difference occurs under the earlier, stronger disruption. Distance-based choices then spend more time queueing and complete fewer trips, while the DQN collects higher-value visits.

The adaptive greedy policy remains competitive. The DQN's mean reward gains over this heuristic range from 0.99 to 1.59 points. Under disruption, the DQN reaches 53.45% preference satisfaction compared with 51.24% for adaptive greedy, but its completion rate is 98.33% compared with 100.00%. The value gain therefore accompanies a small increase in deadline risk.

5.1 Coverage, waiting and paired comparisons

Figure 4 separates the components that contribute to reward. Peak queues reduce coverage and preference satisfaction for every policy. The DQN collects 38.79% of available preference value in this scenario, compared with 32.75% for the nearest policy. Its total queue time is higher, at 89.23 minutes versus 83.61 minutes. More waiting can accompany a more valuable trip when the visitor successfully completes additional attractions.

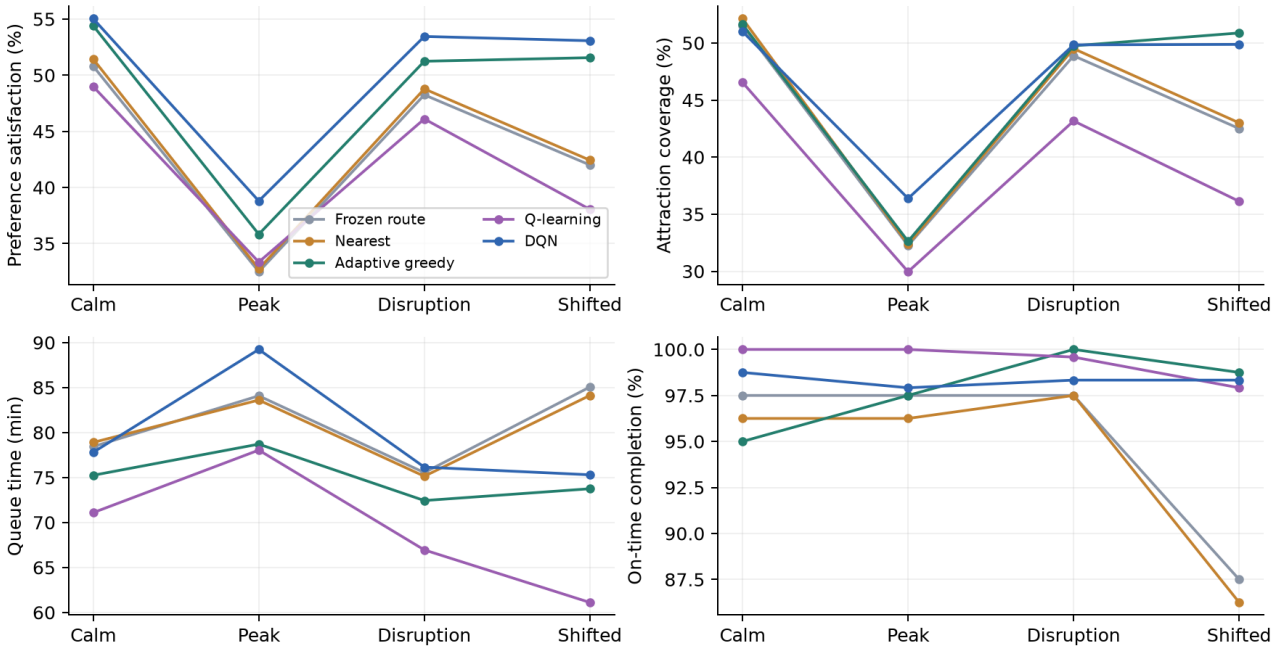


Figure 4: Preference satisfaction, coverage, queue time and on-time completion across the four scenarios. Each point is a mean over 80 cases; learned outcomes also average over three training seeds.

Q-learning completes trips reliably at the main budget, but its lower coverage and collected value produce lower reward than the DQN. The table representation distinguishes visited sets and time bins without sharing estimates between similar states. The DQN shares parameters across observations and uses the full current queue and travel vectors. These implementation differences provide plausible explanations for the observed gap, although the experiment does not isolate their individual effects.

For each case, I average each learner's reward across training seeds and subtract the alternative policy's reward. A paired bootstrap resamples the 80 case differences 2,000 times and uses the 2.5th and 97.5th percentiles as an interval. Figure 5 shows these comparisons. The intervals characterize variation across cases conditional on the three trained models; they do not represent uncertainty across all possible training runs.

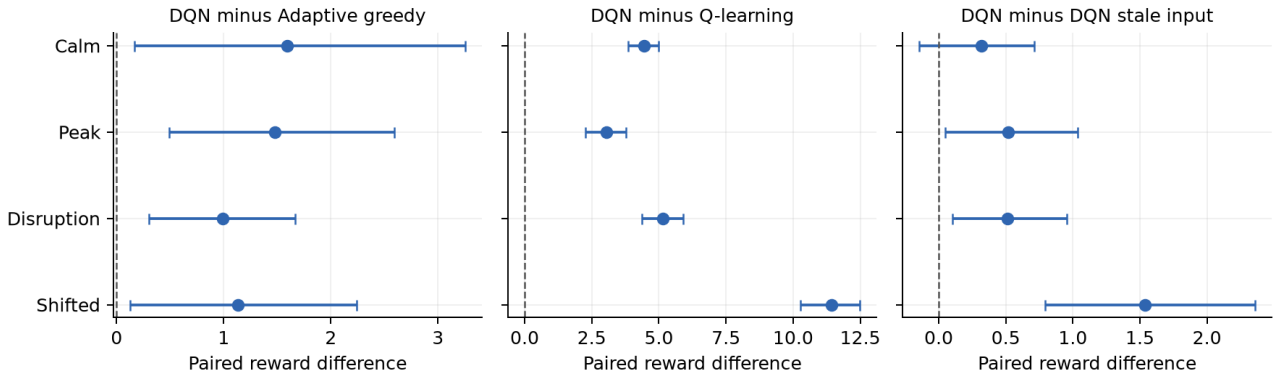


Figure 5: DQN reward differences with paired 95% bootstrap intervals. Positive differences favour the DQN. The stale-input comparison keeps the current feasibility mask available to both policies.

The DQN–adaptive-greedy intervals remain above zero in the four tested scenarios, but their estimated advantages are small. The live-input advantage is clearest in shifted conditions: 1.54 points with interval [0.79, 2.36]. The calm-scenario interval for live versus stale input includes zero. The evidence therefore supports a scenario-

dependent benefit from updated queue and travel features rather than a uniform contribution in every setting.

5.2 Time budgets and an individual route

The budget experiment uses 40 additional disruption cases at 120, 150 and 180 minutes, with no retraining. Figure 6 shows that collected preference value generally rises with more available time for the adaptive heuristic and DQN. At 120 minutes, however, the DQN collects 40.18% preference value and completes 85.00% of trips. Adaptive greedy obtains 41.56% and 92.50%, respectively. The learner's ranking at the training budget does not extend to every time constraint.

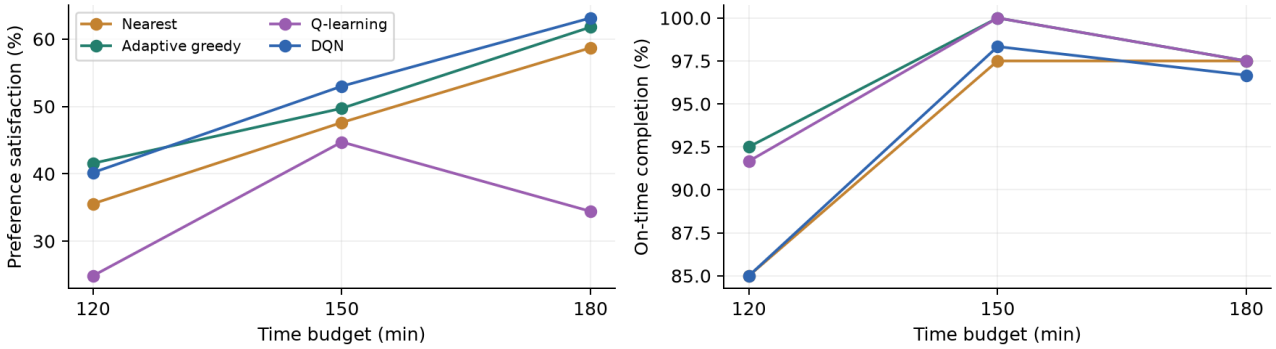


Figure 6: Budget sensitivity on 40 additional disruption cases. All learners were trained at 150 minutes. The Q-learning state key contains a budget bin, so the 120- and 180-minute tests encounter previously unseen budget keys.

The Q-learning budget curve requires special care. Its categorical budget key differs at 120 and 180 minutes, so these cases begin with unlearned table entries and zero action values. Seeded tie breaking then influences the resulting route. The curve reflects this representation's lack of cross-budget sharing and cannot establish a general weakness of Q-learning under longer visits. The DQN also faces a changed input distribution, but its shared network can reuse some learned relationships.

Figure 7 follows one fixed disruption case and the DQN model from training seed 11. Both policies complete five attractions and reach the exit on time. Their attraction sequences differ: adaptive greedy visits labels 2, 3, 7, 6, 5, while the DQN visits 9, 10, 7, 6, 2. The DQN collects 55.32% of preference value compared with 52.20% for adaptive greedy, and their rewards are 39.17 and 37.36.

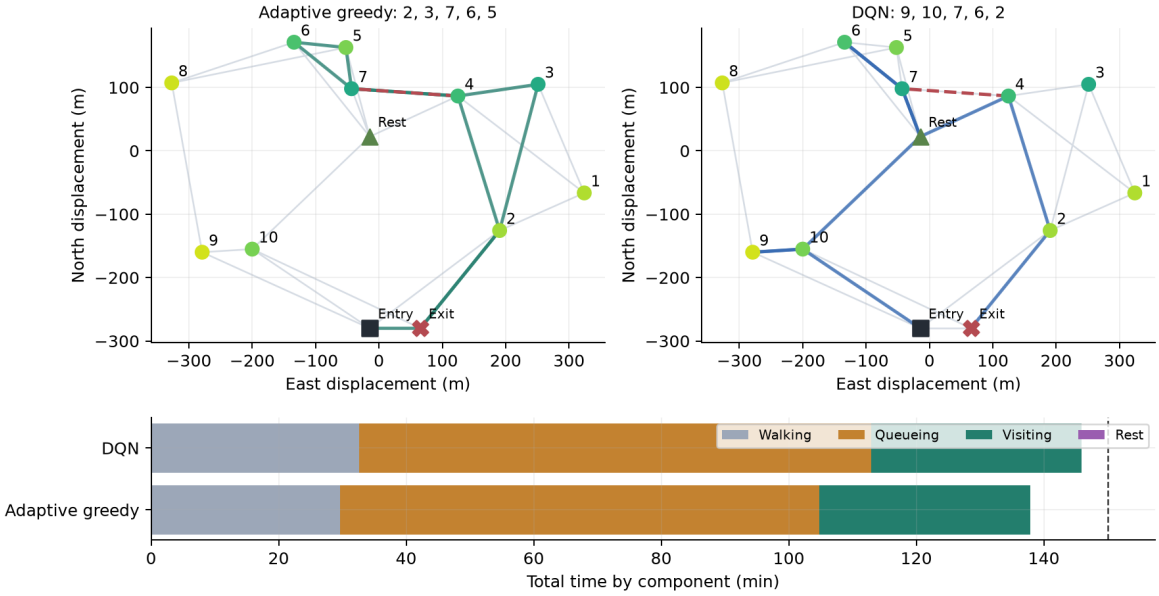


Figure 7: Two routes for a fixed disruption case. The dashed red edge is the road affected during the disruption window. The bars decompose elapsed time; the vertical dashed line marks the 150-minute budget.

The DQN finishes at 145.95 minutes, compared with 137.83 minutes for adaptive greedy. Its queue and walking times are 80.30 and 32.66 minutes, compared with 75.22 and 29.61 minutes for adaptive greedy. Its higher value leaves a smaller exit margin. This example illustrates how two routes with equal coverage can differ in preference

satisfaction and deadline exposure; the case does not determine the aggregate conclusion.

6 Result Interpretation

Results vary moderately across training seeds (Table 3); DQN reward in the peak scenario ranges from 23.40 to 24.90. Together with Figure 3, this variation provides no evidence of convergence to an optimal policy.

Table 3: Mean test reward by training seed, averaged over 80 cases.

Learner	Seed	Calm	Peak	Disruption	Shifted
Q-learning	11	36.07	21.71	32.52	26.86
Q-learning	22	36.44	20.57	33.12	26.58
Q-learning	33	33.16	21.69	33.22	25.95
DQN	11	39.45	23.40	37.86	38.24
DQN	22	40.25	24.81	38.26	38.18
DQN	33	39.31	24.90	38.15	37.24

The DQN achieves the highest mean reward across the four main scenarios. Distance-based policies generally walk less but collect less preference value, while adaptive greedy captures much of the benefit of current information. The stale-input results suggest that updated queue and travel estimates are particularly useful when disruption timing changes.

Performance also depends on the time budget and state representation. At 120 minutes, the DQN makes less reliable exit decisions, while Q-learning encounters unseen budget states. These findings apply to the specified ten-attraction environment and reward weights. Adaptive greedy remains competitive under tighter budgets, and the coverage, preference and completion measures clarify tradeoffs that reward alone may conceal.

References

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